Lesson 16: Practical Considerations in Experiments

English Stage 4 (Psychology) · Unit 3: Research Designs

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 5.5: Practical Considerations; 5.6: Experimental Research (Summary).

In this lesson

You will learn the practical steps that turn a design into a real experiment: recruiting and selecting participants, standardizing the procedure, keeping records, checking the manipulation and pilot testing. You will practise the language of procedural advice and turn it into the future passive of a research proposal.

Learning goals

After this lesson you should be able to:

- Describe several ways to recruit participants and explain how volunteers differ from non-volunteers.
- Explain why a procedure must be standardized, and give several ways to do it, including blinding.
- Explain the purpose of record keeping, manipulation checks and pilot tests.
- Turn procedural advice into the future passive of a research proposal.

Key language

Term or phrase	Meaning	Example
subject pool	a group of people who have agreed to be contacted about taking part in research	introductory psychology students who must take part in studies for a course requirement
confederate	a researcher who pretends to be someone else, often another participant	A confederate dropped some computer diskettes.
experimenter expectancy effect	the effect of the researcher's expectations on participants' behaviour	"maze-bright" and "maze-dull" rats
protocol	here, a written document that specifies everything the experimenters do and say	from greeting participants to dismissing them
double-blind / single-blind	neither the participants nor the experimenters know the condition / only the participants do not know	a drug study with a placebo
demand characteristics	features of a study that cue participants about how the researcher expects them to behave	guessing the hypothesis

Term or phrase	Meaning	Example
manipulation check	a separate measure that confirms that the IV was successfully manipulated	a stress questionnaire after a speech threat
pilot test	a small-scale study to make sure a new procedure works as planned	testing friends or classmates first
it is essential to	it is absolutely necessary to	It is essential to keep good records.
it is (always) a good idea to	recommended practice	It is always a good idea to conduct a pilot test.
be wise to	it is sensible for someone to	Researchers are wise to include a manipulation check in a pilot test.

Rules and explanations

The ideas

Recruiting and selecting participants

- **Ways to recruit:** a formal **subject pool**; advertisements; personal appeals to groups that represent the population (for example, speaking at a meeting of residents of a retirement community). Design a study only for populations you can actually reach.
- **Box "The Volunteer Subject":** even paid or credited participants are volunteers, and volunteers differ on average from non-volunteers (Rosenthal & Rosnow, 1976). They are more interested in the topic, more educated, higher in need for approval, higher in IQ, more sociable and higher in social class. This matters for **external validity** if these characteristics change behaviour: a rational argument might persuade volunteers more than the general population.
- **Selecting participants in field experiments:** Guéguen and de Gail (2003) studied smiling and helping in a supermarket. A confederate on a stairway smiled or did not smile at a shopper; shortly afterwards another confederate dropped computer diskettes, and the DV was whether the shopper helped. The IRB would have had to make sure that doing without informed consent was acceptable (no expected harm; ordinary activities). Selection followed a rule set in advance: gaze at the first person who appeared to be 20 to 50 years old, and include that person only if they gazed back. Without a rule, the confederate might, even unintentionally, choose friendly-looking shoppers when set to smile.

Standardizing the procedure

Small differences in the procedure (clear or vague instructions, a warm or cold greeting) add noise; if they differ across conditions, they become **confounding variables**.

- **Box "Experimenter's Sex as an Extraneous Variable":** in a pain study with icy water, male participants tolerated the pain longer with a female experimenter, and female participants longer with a male experimenter (Ibolya, Brake, & Voss, 2004).

- **Experimenter expectancy effect:** Rosenthal and Fode (1963) told students that their genetically similar rats were "maze-bright" or "maze-dull". Over five days the "bright" rats made more correct responses, responded more quickly and improved more steadily. Students who expected fast learning felt more positively about their rats and reported handling them in a friendlier way.

Ways to standardize:

1. A written protocol for everything experimenters do and say.
2. Standard instructions, read by participants or read to them word for word.
3. Automation of the rest of the procedure (software, slide shows).
4. Anticipated questions, with standard answers.
5. Training of all experimenters together, practising on each other.
6. Every experimenter tests participants in all conditions.

Blinding reduces expectancy effects: in a **double-blind study** neither participants nor experimenters know the condition; in a **single-blind study** only participants are blind. Blinding is impossible when you are both the investigator and the only experimenter, or when the experimenter must carry out the conditions differently.

Records, manipulation checks and pilot tests

- **Record keeping:** prepare the sequence of conditions in advance; add demographics, the date, time and place, the experimenter's name and comments on unusual events. Give each participant an identification number instead of a name, to protect confidentiality.
- **Manipulation check:** when the IV is a construct manipulated indirectly (stress, through the threat of giving a speech), measure it separately (a stress questionnaire, blood pressure). It is most important when results are **null**: if happy or sad music has no effect on recall, a mood measure shows whether mood truly has no effect or the music failed to change mood. Checks usually come at the end of the procedure, to see whether the effect lasted and to avoid demand characteristics.
- **Pilot test:** a small study with formally or informally recruited participants. Do participants understand the instructions? Do they become bored? Does an indirect manipulation work? Can they guess the hypothesis? How long does it take? Are data recorded correctly? Tell pilot participants that you genuinely want criticism.

Chapter 5 in brief (Section 5.6)

The Key Takeaways repeat the chapter: manipulation, measurement and control; random assignment or counterbalancing against confounds; internal and external validity; recruitment, standardization and pilot testing.

Language focus

Giving procedural advice. The section advises the reader with imperatives and evaluative frames:

- Imperatives: "**Create** a written protocol ..."; "**Be sure** that each experimenter tests participants in all conditions."
- *It is* + adjective + *to*: "**It is essential to** keep good records"; "**It is always a good idea to** conduct a pilot test."
- *It is* + adjective + *that* + base verb (the subjunctive): "It is extremely important **that** this kind of selection **be done** according to a well-defined set of rules."

A proposal turns this advice into plans. Use the future passive when the actor is unimportant: "A written protocol **will be created**." Use the active when the actor matters: "Each experimenter **will test** participants in all conditions."

> **Note for Arabic speakers** - In "it is important that this selection **be** done", formal English uses the base form *be*, not *is*, after *important that*, *essential that* and *require that* (British English also accepts *should be*).

Practice

1. Reading questions

1. Name three ways to recruit participants.
2. List four ways in which volunteers differ from non-volunteers. When is this a problem?
3. In Guéguen and de Gail's study, what was the selection rule, and why was a rule necessary?
4. What did Ibolya, Brake and Voss (2004) find?
5. In Rosenthal and Fode's study, what explained the difference between the "maze-bright" and "maze-dull" rats?
6. What is the difference between a single-blind and a double-blind study? When is blinding impossible?
7. Why should participants be identified by numbers rather than names?
8. Why are manipulation checks especially important when results are null, and why are they usually done at the end?

2. Book exercises

From the chapter exercises in Section 5.6. (Exercises 1-4 were done in Lessons 13 and 14.)

1. Practice: List two ways that you might recruit participants from each of the following populations: (a) elderly adults; (b) unemployed people; (c) regular exercisers; (d) math majors.
2. Discussion: Imagine a study in which you will visually present participants with a list of 20 words, one at a time, wait for a short time, and then ask them to recall as many of the words as they can. In the stressed condition, they are told that they might also be chosen to give a short speech in front of a small audience. In the unstressed condition, they are not told that they might have to give a speech. What are several specific things that you could do to standardize the procedure?

3. Language exercise

Rewrite the book's advice as sentences for a research proposal. Use the future passive where the actor is unimportant.

1. Create standard instructions that participants read themselves.
2. Automate the rest of the procedure as much as possible.
3. Train multiple experimenters on the protocol together.
4. Assign an identification number to each participant.
5. Conduct a pilot test of your experiment.

Speak with your tutor

1. Explain to your tutor how an experimenter expectancy effect could occur in your sleep and memory study, and how you will prevent it.
2. In PSYC 110 Lecture 8, Professor Bloom describes Steve Ceci's "Sam Stone" study and notes that the interviewer "was herself unaware of what happened". Discuss why this detail matters, using the term *blind*.
3. PSYC 110 Lecture 20 opens with a reminder about the students' "experimental participation requirement". What kind of recruitment is this, and how might the volunteer characteristics in this lesson affect the results of such studies?
4. Ask your tutor to act as a pilot participant in your study, and interview the "participant" with questions from the pilot-test list.

Writing task

Research project: the full design of your experiment (the last step of Lessons 13-16).

Combine and revise your paragraphs from Lessons 13-15 into a draft "Design and Procedure" section of 350-450 words. Add:

- **participants:** the population, how you will recruit them (subject pool, advertisements or personal appeals) and any rules for including or excluding people;
- **standardization:** at least three methods from the list, and how you will keep the person who tests memory blind to the condition;
- **records:** identification numbers and what you will record for each session;
- **manipulation check:** how you will confirm that the manipulation worked (for example, minutes asleep recorded by a sleep-tracking wristband, or a short questionnaire on whether participants actually slept during the nap);
- **pilot test:** who will take part and three questions it will answer.

Language: future passive for procedures, active voice where the actor matters, one or two evaluative frames (*It is essential that …*).

If you work on your own topic, write the same section for your own experiment.

Self-editing checklist:

- The IV, conditions, DV and design (between or within) are stated clearly.
- Random assignment or counterbalancing is described.
- The recruitment method fits a population you can actually reach.
- At least three standardization methods are included, plus blinding.
- The manipulation check measures the IV, not the DV.
- The pilot test has a clear purpose.
- Future passives are complete (*will be* + past participle).

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- "Study" (xkcd comic printed in the section): http://imgs.xkcd.com/comics/study.png
- "Placebo Blocker" (xkcd comic printed in the section): http://imgs.xkcd.com/comics/placebo_blocker.png
- PSYC 110 Lecture 8 (the Sam Stone study and the power of suggestion): https://oyc.yale.edu/psychology/psyc-110/lecture-8

Answer key

1. Reading questions

1. A formal subject pool, advertisements, and personal appeals to groups that represent the population ("Recruiting Participants").
2. Any four: more interested in the topic, more educated, greater need for approval, higher IQ, more sociable, higher social class. It is an external-validity problem when these characteristics are likely to change behaviour, as with persuasion by a rational argument (box "The Volunteer Subject").
3. Gaze at the first person who appeared to be between 20 and 50; that person became a participant only if they gazed back. The rule avoided biased selection, such as choosing friendly-looking shoppers in the smile condition (paragraph after the box).
4. Male participants tolerated the icy-water pain longer when the experimenter was a woman, and female participants longer when the experimenter was a man (box "Experimenter's Sex ...").
5. The students' expectations: those who expected fast learning felt more positively about their rats and reported handling them in a more friendly manner ("Standardizing the Procedure", Rosenthal paragraph).
6. Double-blind: neither participants nor experimenters know the condition; single-blind: only the participants do not know. Blinding is impossible when you are the investigator and the only experimenter, or when the experimenter must run the conditions differently ("Standardizing the Procedure", last paragraph).
7. Identities should be kept as confidential (or anonymous) as possible; numbers also keep each participant's response sheets together ("Record Keeping").
8. A null result could mean that the IV has no effect or that the manipulation failed; the check shows which. At the end, it confirms that the effect lasted and avoids drawing attention to the manipulation, a demand characteristic ("Manipulation Check").

2. Book exercises

1. (example answers; many are possible) (a) Speak at a meeting of residents of a retirement community; advertise in a seniors' centre. (b) Advertise at employment offices; make a personal appeal to a job-seekers' support group. (c) Put up advertisements in gyms; speak to running clubs. (d) Make an announcement in math classes, with the instructor's permission; post advertisements in the math department.
2. (example answer) Script everything from greeting to dismissal in a protocol; give identical on-screen instructions except for the sentence about the speech; present the 20 words by computer for the same time each, with a delay of fixed length; give everyone the same recall time and answer sheet; prepare a standard answer to "Will I really give a speech?"; train experimenters together and have each test both conditions; let the computer deliver the speech instruction so that the experimenter stays blind; add a stress questionnaire at the end as a manipulation check; pilot test the procedure.

3. Language exercise

(example answers)

1. Standard instructions will be created, and participants will read them on their own.
2. The rest of the procedure will be automated as far as possible.
3. All experimenters will be trained on the protocol together.
4. Each participant will be assigned an identification number.
5. The experiment will be pilot tested before data collection begins.

My notes